

Part A: Understanding Users

```
Session Actions Edit View Help
[~] (sneha@kali) [-]
$ whoami
sneha
[~] (sneha@kali) [-]
$ id
uid=1000(sneha) gid=1000(sneha) groups=1000(sneha),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),44(video),46(plugdev),100(users),101(netdev),102(scanner),112(bluetooth),117(lpdn),119(wireshark),123(vboxsf),124(kdboxer)
[~] (sneha@kali) [-]
$ cat /etc/passwd
root:x:0:0:root:/root:/usr/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lpix:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:11:11:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:mailing list Manager:/var/list:/usr/sbin/nologin
lirc:x:39:39:lirc:/run/lirc:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/usr/sbin/nologin
dhcpd:x:996:996:DHCP Client Daemon:/usr/lib/dhcpd:/bin/false
lss:x:998:998:lss user for lsm2:/usr/sbin/nologin
strongswan:x:100:65534::/var/lib/strongswan:/usr/sbin/nologin
systemd-timesync:x:989:989:systemd Time Synchronization:/usr/sbin/nologin
gnopish:x:101:104::/var/lib/gnupsh:/usr/sbin/nologin
iodine:x:102:65534::/run/iodine:/usr/sbin/nologin
messagbus:x:988:988:system Message Bus:/nonexistent:/usr/sbin/nologin
tcpdump:x:987:987:tcpdump:/nonexistent:/usr/sbin/nologin
niredo:x:103:65534::/var/run/niredo:/usr/sbin/nologin
rpc:x:104:65534::/run/rpcbind:/usr/sbin/nologin
statd:x:105:65534::/var/lib/dfs:/usr/sbin/nologin
redis:x:106:106::/var/lib/redis:/usr/sbin/nologin
mosquitto:x:107:109::/var/lib/mosquitto:/usr/sbin/nologin
redisearch:x:108:108::/var/run/redisearch:/usr/sbin/nologin
stunnel4:x:985:985:stunnel service system account:/var/run/stunnel4:/usr/sbin/nologin
sshd:x:984:65534:sshd user:/run/ssh:/usr/sbin/nologin
dnsmasq:x:983:65534:dnsmasq:/var/lib/dnsmasq:/usr/sbin/nologin
Debian-smp:x:109:111::/var/lib/smp:/bin/false
nfsd:x:982:982:Nfsd server:/nonexistent:/bin/false
sblx:x:110:113::/nonexistent:/usr/sbin/nologin
```

1. What is your current username?

Ans: The username shown by the `whoami` command is sneha.

2. What is UID?

Ans: UID (User ID) is a unique number assigned to a user account in Linux. It identifies the user to the operating system.

Here it is
`uid=1000(sneha)`

3. What is GID?

Ans: GID (Group ID) is a unique number assigned to a group. It identifies the group to which a user belongs.

Here it is
`gid=1000(sneha)`

4. What information does `/etc/passwd` contain?

Ans: The `/etc/passwd` file stores information about user accounts, such as: Username, User ID (UID), Group ID (GID), User description, Home directory, Default login shell.

`sneha:x:1000:1000:sneha ,,:/home/sneha:/bin/bash`

This entry tells us:

- **sneha** → Username
- **x** → Password placeholder (actual password is stored elsewhere)

- **1000** → UID (User ID)
- **1000** → GID (Group ID)
- **Sneha ,,,:** → User description
- **/home/sneha** → Home directory
- **/bin/bash** → Default shell

Part B: Create Users & Groups

```

[~] Session Actions Edit View Help
[~] sneha@kali: ~
[~] $ sudo groupadd interns
[sudo] password for sneha:
[~] sneha@kali: ~
[~] $ sudo groupadd cyberteam
[~] sneha@kali: ~
[~] $ sudo useradd student1
[~] sneha@kali: ~
[~] $ sudo useradd student2
[~] sneha@kali: ~
[~] $ sudo useradd student3
[~] sneha@kali: ~
[~] $ sudo usermod -aG interns student1
[~] sneha@kali: ~
[~] $ sudo usermod -aG cyberteam student2
[~] sneha@kali: ~
[~] $ sudo usermod -aG interns student3
[~] sneha@kali: ~
[~] $ group student1
student1 : student1 interns
[~] sneha@kali: ~
[~] $ group student2
student2 : student2 cyberteam
[~] sneha@kali: ~
[~] $ group student3
student3 : student3 interns
[~] sneha@kali: ~
[~] $ id student1
uid=1001(student1) gid=1001(student1) groups=1001(student1),1001(interns)
[~] sneha@kali: ~
[~] $ id student2
uid=1002(student2) gid=1004(student2) groups=1004(student2),1002(cyberteam)
[~] sneha@kali: ~
[~] $ id student3
uid=1003(student3) gid=1005(student3) groups=1005(student3),1001(interns)
[~] sneha@kali: ~

```

1.To Create groups

sudo groupadd interns
sudo groupadd cyberteam

2.To Create users

sudo useradd student1
sudo useradd student2
sudo useradd student3

3.To Add users to groups

sudo usermod -aG interns student1
sudo usermod -aG cyberteam student2
sudo usermod -aG interns,cyberteam student3

4.Verify group membership

groups student1
groups student2
groups student3

5.For Displaying user and group IDs

id student1
id student2

Part C: File Ownership

- **Original Owner:**Sneha
- **New Owner:**student3
- **Command Used:** sudo chown student3 report.txt
- **Observation:**The ownership of report.txt was successfully changed from sneha to student3. This was verified using the ls -l command.

```
sneha@kali: ~/Cybersecurity_Project
$ mkdir Cybersecurity_Project
$ cd Cybersecurity_Project
$ touch report.txt notes.txt credentials.txt
$ ls
credentials.txt  notes.txt  report.txt
$ ls -l
total 0
-rw-rw-r-- 1 sneha sneha 0 Jun 13 08:49 credentials.txt
-rw-rw-r-- 1 sneha sneha 0 Jun 13 08:49 notes.txt
-rw-rw-r-- 1 sneha sneha 0 Jun 13 08:49 report.txt
$ sudo chown student3 report.txt
[sudo] password for sneha:
$ ls -l
total 0
-rw-rw-r-- 1 sneha sneha 0 Jun 13 08:49 credentials.txt
-rw-rw-r-- 1 sneha sneha 0 Jun 13 08:49 notes.txt
-rw-rw-r-- 1 student3 sneha 0 Jun 13 08:49 report.txt
$
```

Part D: File Permissions

```
sneha@kali: ~/Cybersecurity_Project
$ touch security_policy.txt
$ ls -l security_policy.txt
-rwxrwxrwx 1 sneha sneha 0 Jun 13 09:08 security_policy.txt
$ chmod 444 security_policy.txt
$ ls -l security_policy.txt
-r--r--r-- 1 sneha sneha 0 Jun 13 09:08 security_policy.txt
$ chmod 664 security_policy.txt
$ ls -l security_policy.txt
-rw-rw-r-- 1 sneha sneha 0 Jun 13 09:08 security_policy.txt
$ chmod 777 security_policy.txt
$ ls -l security_policy.txt
-rwxrwxrwx 1 sneha sneha 0 Jun 13 09:08 security_policy.txt
$
```

Commands Used

- **touch security_policy.txt:-** Creates a new empty file named security_policy.txt.
- **ls -l security_policy.txt:-**Displays detailed information about the file, including permissions, owner, group, size, and modification date.
- **chmod 444 security_policy.txt:-** Changes the file permissions to read-only for the owner, group, and others (r--r--r--).
- **chmod 664 security_policy.txt:-**Changes the file permissions so that the owner and group can read and write, while others can only read (rw-rw-r--).

- **chmod 777 security_policy.txt**:-Grants full permissions (read, write, and execute) to the owner, group, and others (rwxrwxrwx).
- **ls -l**:-Used after each permission change to verify that the new permissions have been applied successfully.

Part E: Permission Analysis

755

- **Owner Rights:** Read, Write, Execute (rwx)
- **Group Rights:** Read, Execute (r-x)
- **Other User Rights:** Read, Execute (r-x)
- **Real-world Use Case:** Used for directories and program files where the owner can modify the file, but others can only view and run it.

644

- **Owner Rights:** Read, Write (rw-)
- **Group Rights:** Read (r--)
- **Other User Rights:** Read (r--)
- **Real-world Use Case:** Commonly used for text files, configuration files, and documents that only the owner should edit.

777

- **Owner Rights:** Read, Write, Execute (rwx)
- **Group Rights:** Read, Write, Execute (rwx)
- **Other User Rights:** Read, Write, Execute (rwx)
- **Real-world Use Case:** Gives full access to everyone. Generally avoided because it creates security risks.

600

- **Owner Rights:** Read, Write (rw-)
- **Group Rights:** No permissions (---)
- **Other User Rights:** No permissions (---)
- **Real-world Use Case:** Used for sensitive files such as passwords, private keys, and confidential information.

700

- **Owner Rights:** Read, Write, Execute (rwx)
- **Group Rights:** No permissions (---)
- **Other User Rights:** No permissions (---)
- **Real-world Use Case:** Used for private directories where only the owner should have access.

Part F: Security Challenge

File	Recommended permission	Reason
password_backup.txt	600	Contains sensitive information. Only the owner should be able to

		read and modify it.
public_notice.txt	644	Everyone can read the file, but only the owner can edit it.
system_log.txt	640	The owner can read and write, the group can read, and others have no access. This protects system information.
personal_notes.txt	600	Personal notes should be accessible only to the owner for privacy.

Part G: Linux Security Research

1. Why are file permissions important?

Ans: File permissions help keep files safe from unauthorized users. They control who can read, change, or use a file. This helps protect important information and improves system security.

2. What happens if sensitive files are given 777 permissions?

Ans: Permission 777 gives full access to everyone on the system. Any user can read, modify, or delete the file. This can lead to data theft, misuse, or accidental changes.

3. What is the Principle of Least Privilege?

Ans: The Principle of Least Privilege means giving users only the access they need to do their work. They should not get extra permissions. This reduces security risks and helps protect the system.

4. Why do organizations restrict user access?

Ans: Organizations restrict user access to protect important data and systems. Not every employee needs access to every file. Limiting access helps prevent mistakes, data leaks, and unauthorized actions.